WORKED SOLUTION

STEP II 1988, Question 13

Mechanics • Friction and Equilibrium

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Read the question before you go on. It is in the STEP Support Programme's question database:

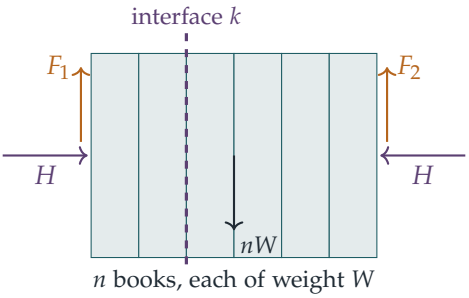
step.maths.org/questions/88-s2-q13

The key idea

Every vertical force that holds the books up is friction, and friction at any contact is capped at μ times the normal reaction there. So we need two things: the normal reaction at every contact (horizontal equilibrium makes this easy) and the friction each contact must supply (vertical equilibrium of a suitable *block* of books). Then we find which contact is closest to slipping. Treating a group of neighbouring books as one body keeps the algebra short.

Solution

Model. Suppose there are n identical books, each of weight W, with their covers vertical. Each hand presses horizontally with force $H \leq Q$ and supplies an upward frictional force; call these F_1 (left hand) and F_2 (right hand). Because each hand's vertical force is exactly this friction, we need $F_1, F_2 \leq P$. The question tells us to ignore turning effects, so we only need force balance on each book or block of books.



Normal reactions. Take the books numbered $1, 2, \ldots, n$ from left to right, and let interface k be the contact between book k and book $k+1$ ($1 \leq k \leq n-1$). The only horizontal forces on the block of books $1, \ldots, k$ are the left hand's push H and the

normal reaction from book $k+1$. So that reaction is also H: *every* contact, hand or book, has normal reaction H, and so every contact can supply friction of size at most μH.

The hands. Vertically, for the whole row,

$$F_1 + F_2 = nW.$$

Each of F_1, F_2 is at most P (arm strength) and at most $\mu H \le \mu Q$ (friction). Write $L = \min(P, \mu Q)$. Then $nW = F_1 + F_2 \le 2L$, so

$$n \le \frac{2L}{W} = \frac{2\min(P, \mu Q)}{W}.$$

This is necessary. If it holds, the best plan is to press as hard as possible, $H = Q$ (a larger H only raises every friction limit), and to share the load equally, $F_1 = F_2 = \frac{1}{2}nW \le L$. Then both hands are within their limits.

> **Ask yourself** Could the books slip past one another somewhere in the middle, even though the hands hold?

The interfaces between books. We check this directly. Apply vertical equilibrium to the block of books $1, \dots, k$. Its weight kW is held up by $F_1 = \frac{1}{2}nW$ from the hand and by a friction force f_k (upwards positive) from book $k+1$:

$$f_k = kW - \tfrac{1}{2}nW = \left(k - \tfrac{n}{2}\right) W.$$

For $1 \le k \le n-1$ we have $\left|k - \frac{n}{2}\right| \le \frac{n}{2} - 1$, with equality at $k = 1$ and $k = n-1$, the interfaces next to the outer books. So the largest friction any interface needs is $\left(\frac{n}{2} - 1\right)W$, which is W less than the $\frac{1}{2}nW$ each hand supplies. Since every interface has the same limit μQ as a hand, if the hands do not slip then no pair of books slips. (Physically: each outer book is held by its hand, and the friction it passes inwards is the hand's friction minus its own weight.)

Answer. The row can be lifted exactly when $nW \le 2\min(P, \mu Q)$. So the maximum number of books is

$$\boxed{n_{\max} = \left\lfloor \frac{2\min(P,\, \mu Q)}{W} \right\rfloor}$$

that is, the largest integer not exceeding both $2P/W$ and $2\mu Q/W$. Which bound applies tells us what limits the librarian: arm strength in lifting when $P < \mu Q$, and grip when $\mu Q < P$.

Checks

- Take $W = 10\,\text{N}$, $P = 60\,\text{N}$, $Q = 100\,\text{N}$ and $\mu = 0.4$. Then $\min(60, 40) = 40$ and $n_{\max} = 8$. With 8 books each hand gives $40\,\text{N}$ of friction, exactly its limit, and the worst interface needs $30\,\text{N} < 40\,\text{N}$. A ninth book would need $45\,\text{N}$ from each hand.

- With one book ($n = 1$) there are no interfaces, and we just need $W \le 2\min(P, \mu Q)$, which is clearly right.
- By computer we solved the full set of inequalities (any $H \le Q$, any split F_1, F_2, every interface) for 3000 random choices of W, P, Q, μ, and the largest feasible n always matched the formula.

Common slips

- Forgetting that the arm's vertical force *is* the friction, so both limits P and μQ apply to the same force.
- Assuming the middle books are the weakest point without checking. The forces between books are smaller than at the hands, not larger.
- Giving $2\min(P, \mu Q)/W$ without the floor: the answer must be a whole number of books.

Beyond STEP

Here friction is *statically indeterminate*: only $F_1 + F_2$ is fixed by equilibrium, and we asked whether *some* split keeps every contact within its limit. Questions like this are small linear programmes, and the same idea of checking whether some equilibrium is admissible underlies the limit analysis of structures. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see friction and equilibrium treated carefully.

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